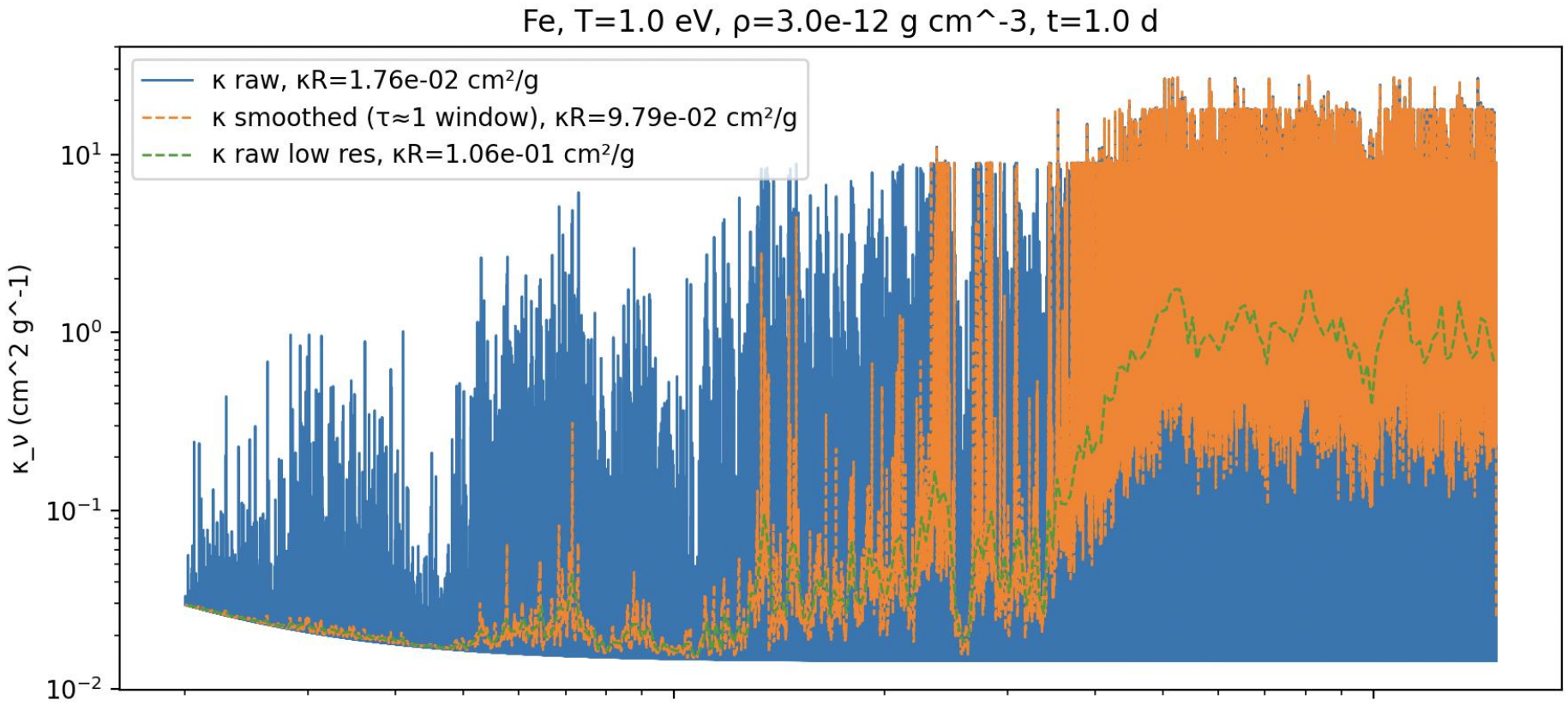
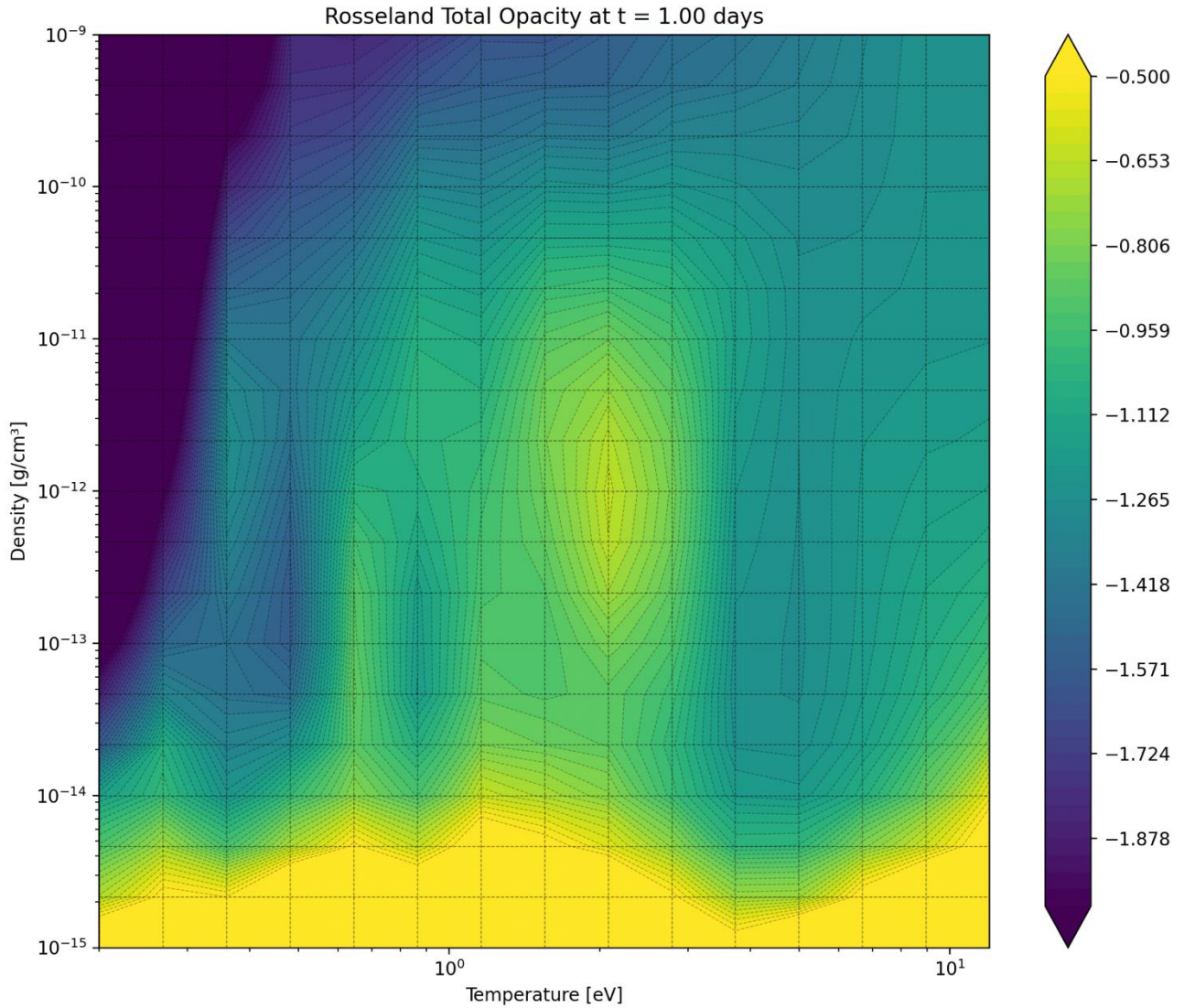
Supernove Gray Opacity modeling

Summery of previous times:

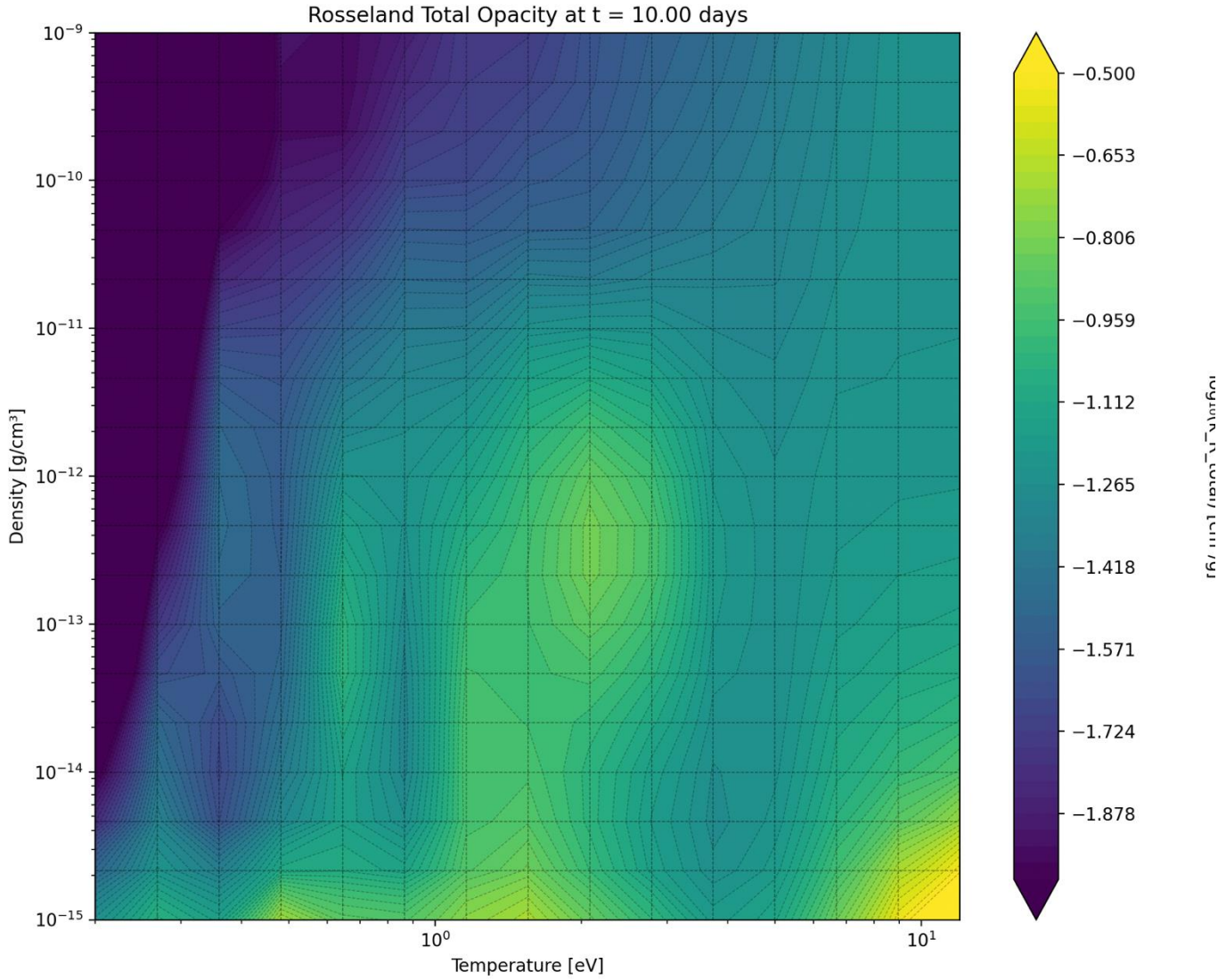
Expansion opacity can be estimated to one number of rosseland mean opacity by practically smoothing with tau=1 window.



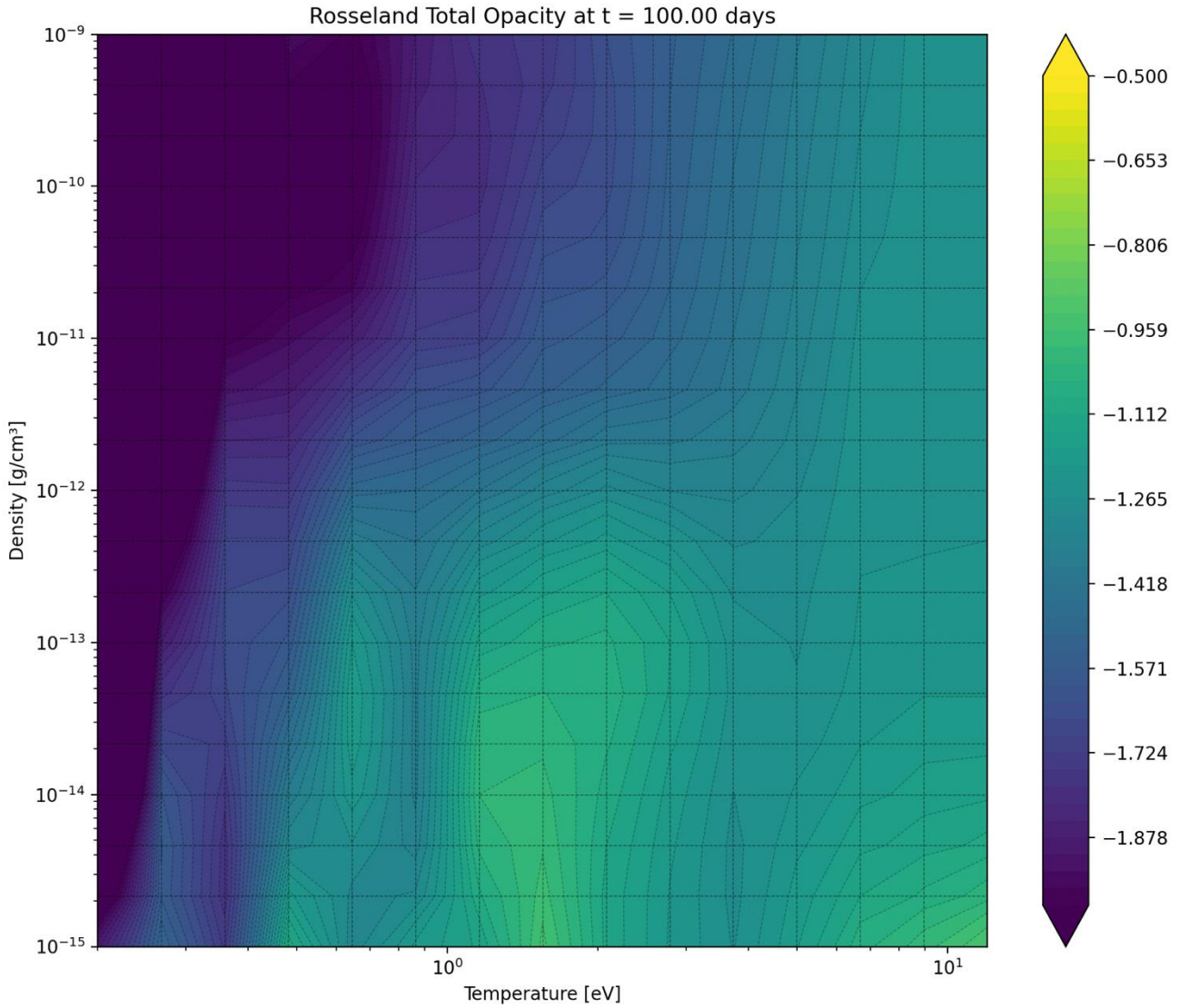
The opacity itself



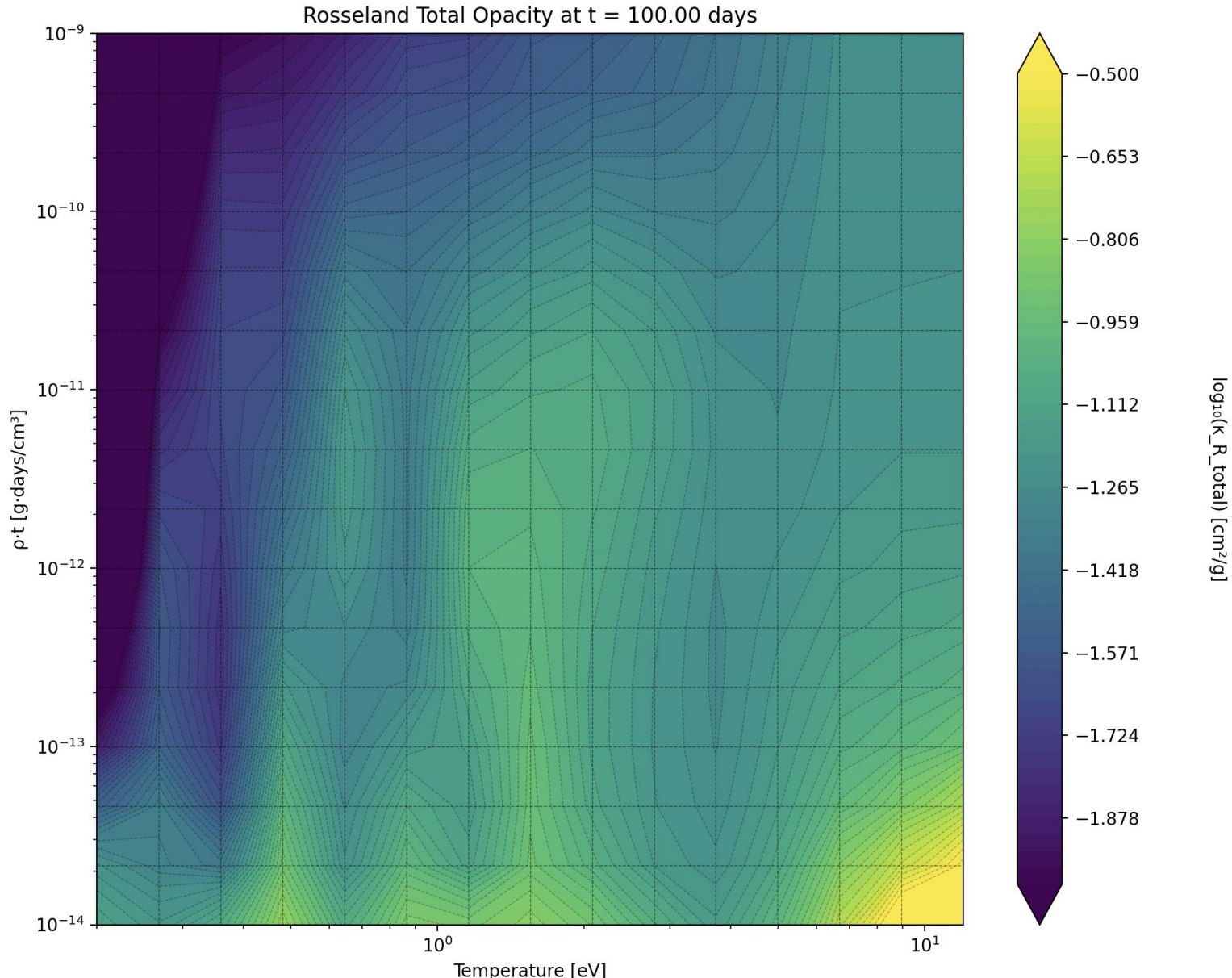
The opacity itself



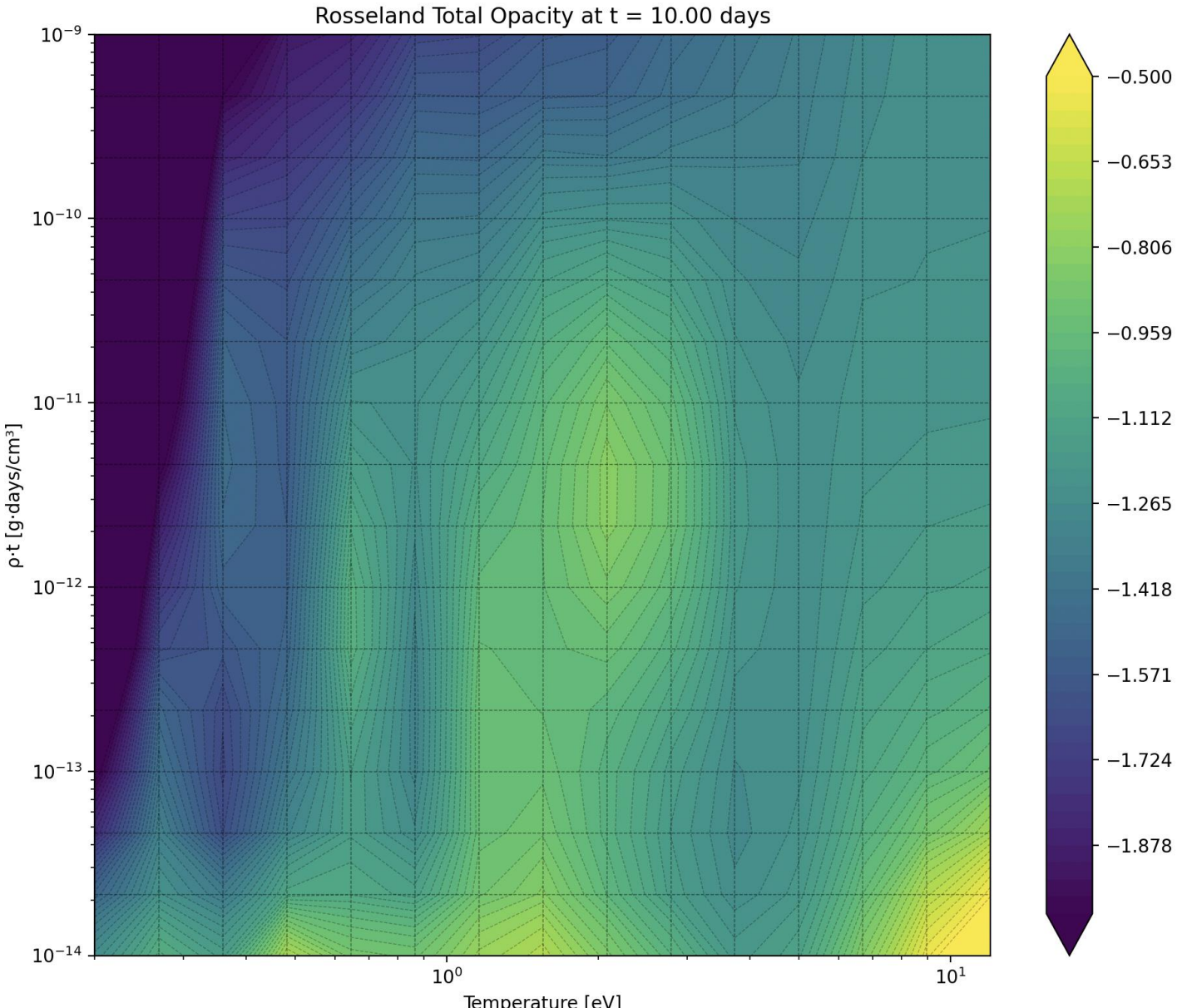
The opacity itself



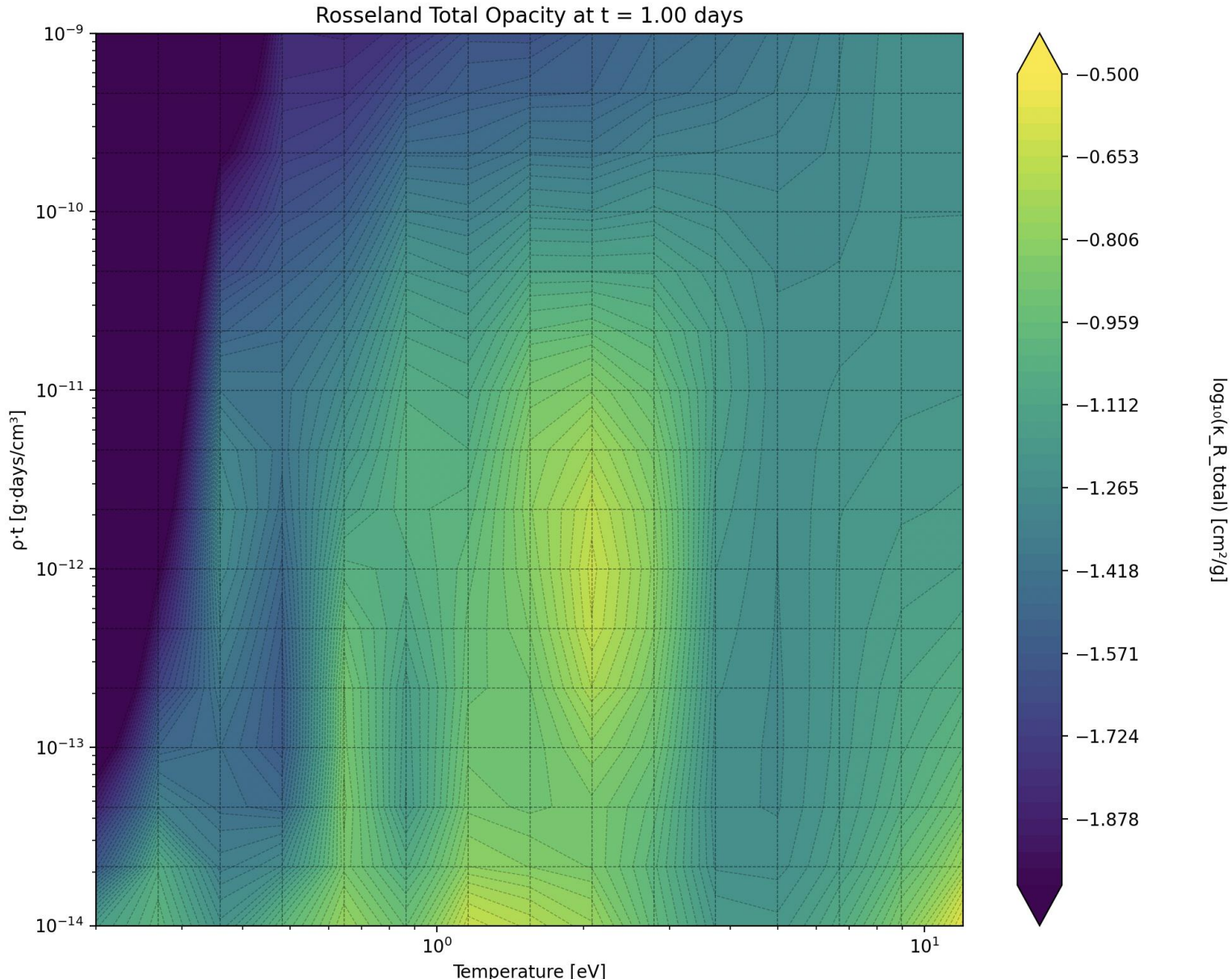
The opacity as a function of (**rho*t** ,T)



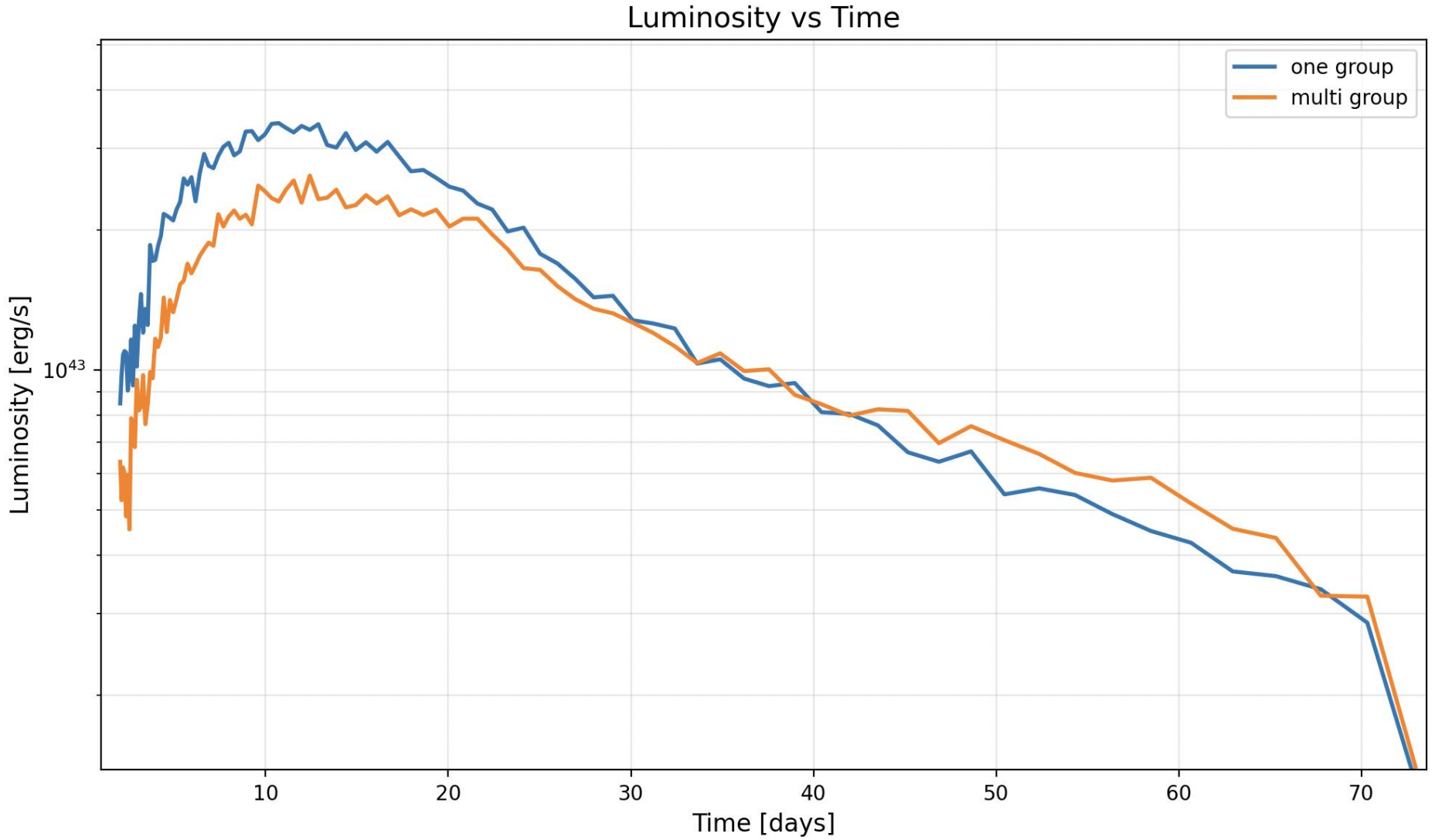
The opacity as a function of (**rho*t** ,T)



The opacity as a function of (**rho*t** ,T)



Results


Luminosity vs Time
one group
multi group
Luminosity [erg/s]
10^{43}
10
20
30
40
50
60
70
Time [days]

Gap source:

In OG Opacity:

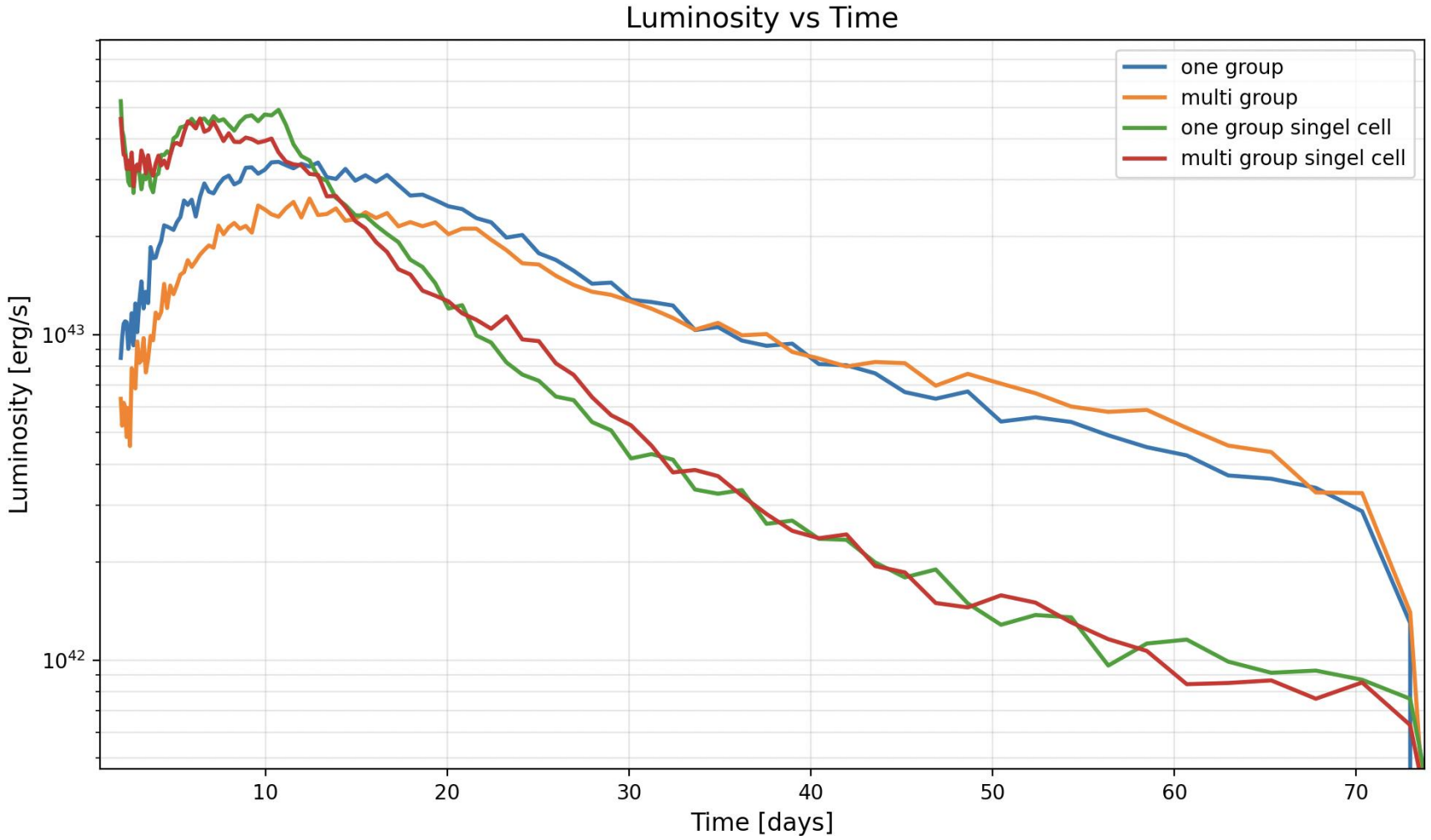
When a photon gets to a new cell with lower temperture he has a **lower chanse** to be absorbed because its frequency gets lower.

In MG Opacity:

When a photon gets to a new cell with lower temperture he has a **similar chanse** to be absorbed.

To test this, an experiment with singel cell and singel Temperture was done

Results


Luminosity vs Time
one group
multi group
one group singel cell
multi group singel cell
Luminosity [erg/s]
10^{43}
10^{42}
10
20
30
40
50
60
70
Time [days]